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PRACTIC

AIM:- VImplement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distributions.

1. Difference between Binomial, Poisson, and Normal Distributions

Distribution	Type	Purpose
Binomial Distribution	Discrete	Models the number of successes in a fixed number of independent trials
Poisson Distribution	Discrete	Models the number of events occurring in a fixed interval of time or space
Normal Distribution	Continuous	Models data that clusters around a mean in a bell-shaped curve

2. Real-World Significance of the Poisson Distribution

The Poisson Distribution is useful when studying rare or random events occurring independently at a constant average rate.

Real-world applications: Number of customer arrivals in a bank per hour Number of phone calls received at a call center Traffic accidents at an intersection per month Number of defects in manufactured products Radioactive decay events in physics Number of emails received in a minute Significance: Helps in queueing theory (used in networks and operations research) Used in risk analysis and reliability engineering Useful in machine learning for modeling count data (e.g., Poisson regression) Assists in forecasting random event frequencies

3. What is the Central Limit Theorem (CLT) and its Relation to Normal Distribution?

The Central Limit Theorem states:

When sufficiently large random samples are taken from any population, the distribution of sample means tends to become normal, regardless of the original population distribution.

Relation to Normal Distribution

As sample size increases:

$X \sim N(\mu, n\sigma)$

This means:

Sample means follow a normal distribution. Many statistical methods assume normality because of CLT. Even non-normal populations can be analyzed using normal approximations when samples are large.

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4. Why Norma

The Normal Di

(a) Many Natu

Examples:

Heights IQ scores Measurement errors Exam marks (b) Supported by Central Limit Theorem

Many averages and aggregated data tend toward normality.

(c) Simplicity and Mathematical Convenience

Its bell curve is defined by:

f(x)= σ 2π

1

e − 2σ 2 (x−μ) 2

sigma mu

This makes calculations easier.

(d) Used in Machine Learning

It appears in:

Gaussian Naive Bayes Linear regression assumptions Error modeling Feature standardization (z-scores) Probabilistic models (e) Basis for Statistical Inference

Confidence intervals, hypothesis testing, and regression often rely on normality.

Double-click (or enter) to edit

5. What is Skewness in Distributions and Which Distributions Exhibit It?

Skewness measures the asymmetry of a probability distribution.

Types of Skewness Positive (Right) Skew Long tail on the right Mean > Median > Mode

Examples:

Income distributions Poisson distribution (when λ is small) Negative (Left) Skew Long tail on the left Mean < Median < Mode

Examples:

Difficult exam scores (many low scores) Zero Skew (Symmetric) Balanced on both sides

Example:

Normal distribu
Skewness Binc
especially for s

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Note:

Binomial becomes more symmetric as n increases and p approaches 0.5. Poisson approaches normal behavior for large λ.

```
# Assignment 7: Probability Distributions in Python

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# -------------------------------
# 1. Binomial Distribution
# -------------------------------
n = 10          # Number of trials
p = 0.5         # Probability of success
x_binom = np.arange(0, n+1)

binom_probs = binom.pmf(x_binom, n, p)

print("Binomial Distribution Probabilities:")
for x, prob in zip(x_binom, binom_probs):
    print(f"P(X={x}) = {prob:.4f}")

plt.figure(figsize=(8,5))
plt.bar(x_binom, binom_probs)
plt.title("Binomial Distribution")
plt.xlabel("Number of Successes")
plt.ylabel("Probability")
plt.show()


# -------------------------------
# 2. Poisson Distribution
# -------------------------------
lam = 4    # Average rate (lambda)
x_poisson = np.arange(0, 15)

poisson_probs = poisson.pmf(x_poisson, lam)

print("\nPoisson Distribution Probabilities:")
for x, prob in zip(x_poisson, poisson_probs):
    print(f"P(X={x}) = {prob:.4f}")

plt.figure(figsize=(8,5))
plt.bar(x_poisson, poisson_probs)
plt.title("Poisson Distribution")
plt.xlabel("Number of Events")
plt.ylabel("Probability")
```

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```
plt.show

# ------
# 3. Nor
# ------
mu = 50          # Mean
sigma = 10       # Standard deviation

x_normal = np.linspace(mu-4*sigma, mu+4*sigma, 1000)
normal_probs = norm.pdf(x_normal, mu, sigma)

plt.figure(figsize=(8,5))
plt.plot(x_normal, normal_probs)
plt.title("Normal Distribution")
plt.xlabel("Values")
plt.ylabel("Density")
plt.show()


# -------------------------------
# 4. Sample Statistics and Skewness Demo
# -------------------------------
# Generate random samples
binom_sample = np.random.binomial(n, p, 1000)
poisson_sample = np.random.poisson(lam, 1000)
normal_sample = np.random.normal(mu, sigma, 1000)

print("\nSample Means:")
print("Binomial Mean =", np.mean(binom_sample))
print("Poisson Mean =", np.mean(poisson_sample))
print("Normal Mean =", np.mean(normal_sample))

print("\nSample Variances:")
print("Binomial Variance =", np.var(binom_sample))
print("Poisson Variance =", np.var(poisson_sample))
print("Normal Variance =", np.var(normal_sample))
```

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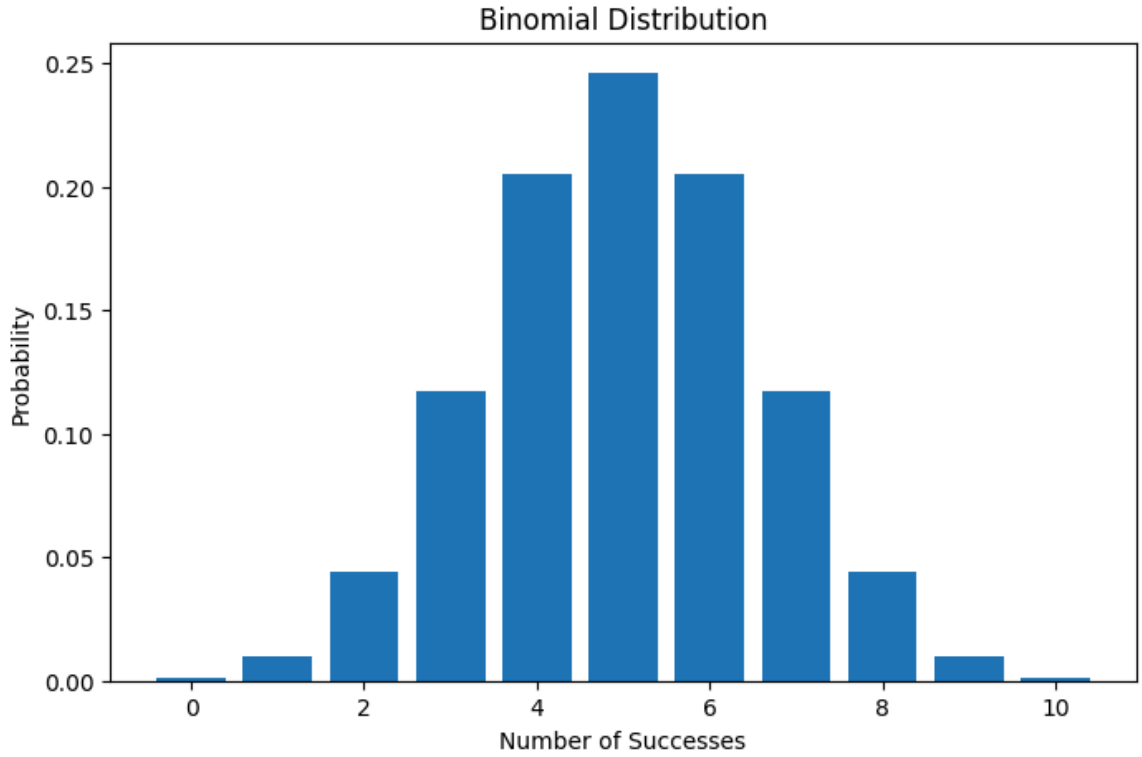
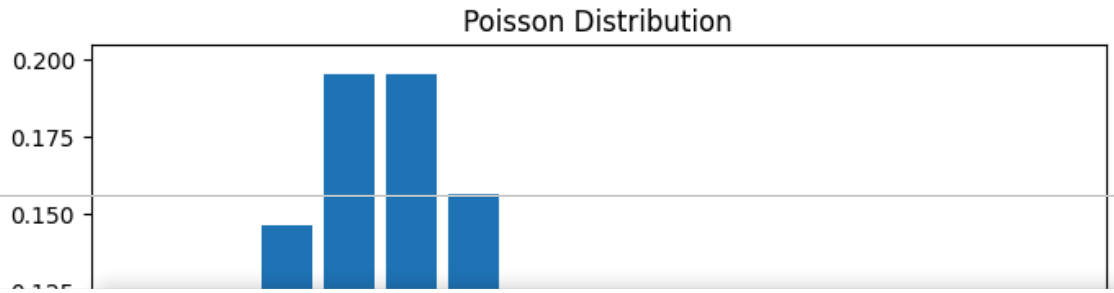
```
Binomial
P(X=0) =
P(X=1) =
P(X=2) =
P(X=3) =
P(X=4) =
P(X=5) =
P(X=6) = 0.2051
P(X=7) = 0.1172
P(X=8) = 0.0439
P(X=9) = 0.0098
P(X=10) = 0.0010
```




```
Poisson Distribution Probabilities:
P(X=0) = 0.0183
P(X=1) = 0.0733
P(X=2) = 0.1465
P(X=3) = 0.1954
P(X=4) = 0.1954
P(X=5) = 0.1563
P(X=6) = 0.1042
P(X=7) = 0.0595
P(X=8) = 0.0298
P(X=9) = 0.0132
P(X=10) = 0.0053
P(X=11) = 0.0019
P(X=12) = 0.0006
P(X=13) = 0.0002
P(X=14) = 0.0001
```



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